

Market Regime-Switching Transformer for Financial Time-Series Forecasting

Course: DATA 612 – Deep Learning

Team Members: Juhi Ramod, Palak Wadhwa, Rohan Jain, Vikranth Reddimasu

1. Background and Significance

Financial markets are highly dynamic and influenced by multiple interrelated factors, including investor sentiment, macroeconomic conditions, and geopolitical events. These forces often lead to different market regimes commonly referred to as bull (rising), bear (falling), or sideways (stable) markets. Traditional time-series forecasting models, such as ARIMA or LSTM networks, assume a single, stationary data-generating process. However, real-world market data are non-stationary and exhibit abrupt regime changes, particularly during crises like the 2008 Global Financial Crash or the 2020 COVID-19 market shock.

Recent advances in Transformer neural networks have demonstrated remarkable success in modeling complex temporal dependencies. Unlike recurrent networks, Transformers rely on self-attention mechanisms that can capture long-range relationships across time. This makes them well-suited for financial data, which often exhibit dependencies across days, weeks, and even months.

The goal of this project is to harness the power of Transformers to both forecast short-term market returns and detect underlying market regimes. By introducing a multi-task learning approach, where the Transformer jointly learns to predict returns and classify latent market states, the model can adapt its internal representation to changing economic conditions. This research has both academic significance as an exploration of regime-adaptive deep learning architectures and practical relevance, offering potential improvements for risk management, algorithmic trading, and economic forecasting.

2. Problem Statement

The financial market alternates between distinct behavioral patterns (regimes) that affect asset price dynamics, volatility, and correlation structures. Most deep learning forecasting approaches treat these changes as noise, leading to poor generalization during regime transitions.

Solution:

Develop a Transformer-based model capable of predicting next-day stock index returns while simultaneously identifying latent market regimes (bull, bear, or neutral). The model should adapt its forecasting behavior based on detected regimes and maintain robustness during volatile periods.

Research Objectives:

1. Design a Market Regime-Switching Transformer that performs both regression (return prediction) and classification (regime detection).
2. Integrate macroeconomic and volatility indicators to enhance the model's situational awareness.
3. Evaluate model performance across multiple time periods, including crisis years, to assess adaptability and stability.
4. Visualize regime transitions and interpret attention mechanisms for explainability.

3. Data Collection and Preparation

We will use publicly available financial and macroeconomic datasets from reliable online repositories:

- Yahoo Finance: S&P 500 (^GSPC), SPY ETF, Nasdaq Index (via yfinance).
- FRED: Volatility Index (VIX), interest rates, inflation (CPI), industrial production (via fredapi).
- Quandl / Kaggle: Additional market indicators such as oil, gold, and USD index.

Feature Engineering:

1. Input Variables: daily closing prices, log returns, trading volume, VIX, interest rates, CPI, yield spreads, and moving averages.
2. Derived Features: rolling volatility, momentum indicators, and lagged returns.
3. Regime Labeling (Weak Supervision): using 20-day rolling return thresholds to define bull, bear, or neutral regimes.
4. Data Cleaning: forward-fill missing data and normalize inputs with z-score scaling.

Dataset Split:

- Training: 2000–2015
- Validation: 2016–2019
- Testing: 2020–2024 (includes COVID crisis)

We will use rolling walk-forward validation to avoid lookahead bias.

4. Model Architecture and Deep Learning Framework

Model Overview:

The Transformer Encoder-Decoder model will have two output heads:

1. Regression Head – predicts next-day returns of the S&P 500 index.
2. Classification Head – predicts the current regime (bull, bear, neutral).

Model Components:

- Input Layer: multivariate time-series window (e.g., 60 previous days × features).
- Embedding Layer: linear projection of input features + positional encoding.
- Transformer Encoder: multi-head self-attention layers to capture long-term temporal dependencies.
- Output Heads: regression (MSE loss) and classification (cross-entropy loss).

Combined Loss Function:

$$L = \lambda_1 \times \text{MSE} + \lambda_2 \times \text{CrossEntropy}$$

Framework:

We will use PyTorch as the main deep learning framework, with supporting libraries such as torchmetrics, scikit-learn, numpy, and matplotlib. All experiments will be run on GPU (Google Colab Pro or CUDA-enabled local machine).

5. Metrics for Validation and Verification

Forecasting Task: MAE, RMSE, Directional Accuracy

Regime Classification: Accuracy, F1-Score, Confusion Matrix

Combined Evaluation: Weighted aggregate metric to balance both tasks

Interpretability: Attention heatmaps and regime transition plots to validate model focus during market shifts.

6. Expected Outcomes and Deliverables

1. A reproducible codebase implementing the Market Regime-Switching Transformer.
2. Baseline comparisons with LSTM, vanilla Transformer, and Hidden Markov Models.
3. A comprehensive report with visualizations of regime transitions, forecasts, and attention heatmaps.
4. Performance tables comparing models across metrics and market periods.

7. References

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